SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
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Section / Batch:	AI & DS	**Date:**	20-07-26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Q1. ***Master Theorem Application for Mixed Polynomial Term***

Consider the recurrence relation $T(n) = 2T(n/2) + n^2 \log n$. Analyze the problem using the Master Theorem and determine the time complexity. Clearly identify parameters and compare f(n) with n^log_b a. Justify the correct case selection and explain reasoning. Use recursion tree diagrams if needed and interpret the result clearly.

Code of Conduct

I __________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Master Theorem Application for Mixed Polynomial Term

NARRAVULA NAVYA SRI

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Artificial Intelligence and Data Science

CSA0614-Design and Analysis of Algorithm

Problem Statement

Objective:

Determine the asymptotic time complexity of the recurrence:

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2 \log n$$

using the **Master Theorem**.

Tasks

- Identify parameters a, b, and $f(n)$.
- Compute $n^{\log_b a}$.
- Compare $f(n)$with $n^{\log_b a}$.
- Select the appropriate Master Theorem case.
- Conclude the time complexity.

Algorithm Overview

The Master Theorem solves recurrences of the form

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

Step 1

Identify:

$$a = 2$$
$$b = 2$$
$$f(n) = n^2 \log n$$

Step 2

Compute

$$n^{\log_b a} = n^{\log_2 2} = n$$

Step 3

Compare

$$f(n) = n^2 \log n$$

with

$$n^{\log_b a} = n$$

Since

$$n^2 \log n = \Theta(n^{1+\epsilon})$$

for $\epsilon = 1$, $f(n)$grows polynomially faster.

Algorithm / Pseudocode

Master Theorem(T(n))

Input:

T(n)=2T(n/2)+n²logn

1. a ⟵ 2
2. b ⟵ 2
3. f(n) ⟵ n²logn
4. Compute n^(log_b a)
5. If f(n)=Θ(n^(log_b a))

   Case 2
6. Else if f(n)= Θ(n^(log_b a−ε))

   Case 1
7. Else if f(n)= Θ(n^(log_b a+ε))

   Check Regularity Condition

   Case 3
8. Output Θ(f(n))

Working Example

Recursion Tree: $n^2 \log n$

```
                 n² log n
                 /      \
(n/2)² log(n/2)   (n/2)² log(n/2)
     /    \          /    \
               ...
```

Cost at each level:

Level 0:

$$n^2 \log n$$

Working Example

Level 1

$$2\left(\frac{n}{2}\right)^2 \log\left(\frac{n}{2}\right) = \frac{n^2}{2}(\log n - 1)$$

Level 2

$$4\left(\frac{n}{4}\right)^2 \log\left(\frac{n}{4}\right) = \frac{n^2}{4}(\log n - 2)$$

General level i

$$\frac{n^2}{2^i}(\log n - i)$$

The cost decreases geometrically.

Total work:

$$n^2 \log n + \frac{n^2}{2}\log n + \frac{n^2}{4}\log n + \cdots = \Theta(n^2 \log n)$$

Complexity Analysis

Parameters:

Parameter	Value
(a)	2
(b)	2
(f(n))	(n^2 log n)

Compute

$$n^{\log_b a} = n$$

Comparison

$$n^2 \log n > n^{1+\epsilon}$$

Choose

$$\epsilon = 1$$

Regularity Condition

Check

$$af(n/b) \leq cf(n)$$

$$2\left(\frac{n}{2}\right)^2 \log\left(\frac{n}{2}\right) = \frac{n^2}{2}(\log n - 1)$$

Complexity Analysis

For sufficiently large n,

$$\frac{n^2}{2}(\log n - 1) \leq c(n^2 \log n)$$

for some constant $c < 1$(e.g., $c = \frac{3}{4}$).

Hence the regularity condition holds.

Master Theorem Case

Case 3

Therefore,

$$\boxed{T(n) = \Theta(n^2 \log n)}$$

Comparative Analysis

Case	Condition	Complexity
Case 1	f(n) smaller	Θ(n^{\log_ba})
Case 2	Same order	Θ(n^{\log_ba}\log^{k+1}n)
Case 3	f(n) polynomially larger	Θ(f(n))

For this recurrence:

$$n^2 \log n \gg n$$

Hence **Case 3** applies.

Applications

Master Theorem is widely used in analysing:

- Divide and Conquer algorithms
- Merge Sort
- Binary Search
- Strassen Matrix Multiplication
- Closest Pair of Points
- Karatsuba Multiplication
- FFT Algorithms
- Computational Geometry algorithms

Results & Discussion

Result
Given
$$T(n) = 2T(n/2) + n^2 \log n$$
we obtain
$$a = 2$$
$$b = 2$$
$$n^{\log_b a} = n$$
$$f(n) = n^2 \log n$$
Since
$$f(n) = \Omega(n^{1+\epsilon})$$
and the regularity condition is satisfied,
Master Theorem Case 3 applies.
Final Complexity
$$\boxed{\Theta(n^2 \log n)}$$

The recursion tree also confirms that the work is dominated by the root level, with subsequent levels decreasing geometrically.

Conclusion & References

Conclusion:

Identified:
$$a = 2$$
$$b = 2$$
$$f(n) = n^2 \log n$$
Computed:
$$n^{\log_b a} = n$$
Compared:

$n^2 \log n$ grows polynomially faster than n

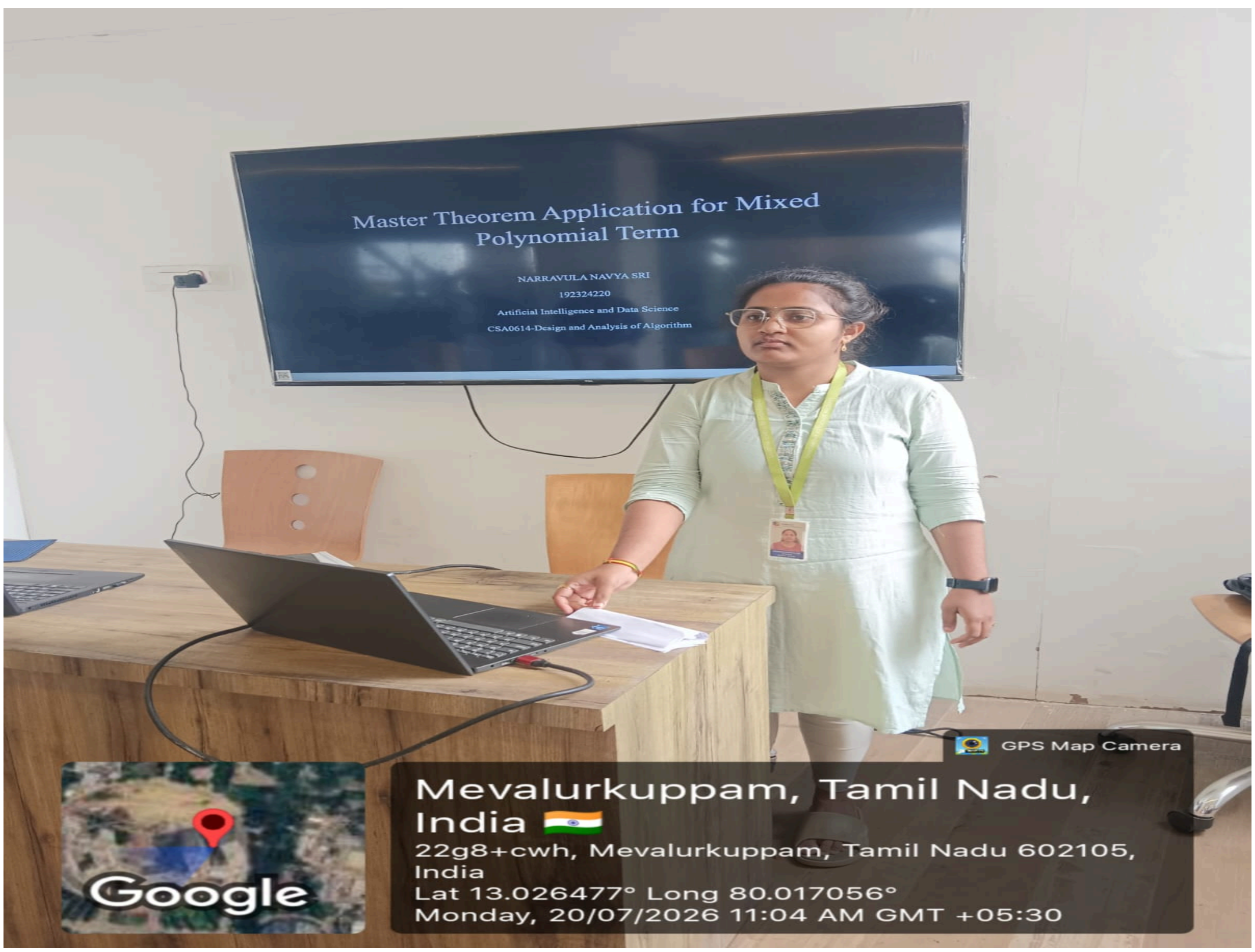
Applied **Master Theorem Case 3**

Final result:
$$\boxed{T(n) = \Theta(n^2 \log n)}$$

References:

- Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, **Introduction to Algorithms (CLRS)**, 4th Edition.
- Anany Levitin, **Introduction to the Design & Analysis of Algorithms**, 4th Edition.
- Michael T. Goodrich, Roberto Tamassia, **Algorithm Design and Applications**.


THANK YOU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